

Decision Log

Skylark Drones — Monday.com Business Intelligence Agent

Technical Assignment · AI Engineer Role

1. Problem Framing & Scope

The core challenge is converting ambiguous, founder-level natural language queries into precise analytics over two intentionally messy datasets — a deal pipeline and a work order tracker — without any preprocessing or caching at query time. The agent must handle both linguistic ambiguity (what does 'energy sector' mean?) and data quality issues (null dates, text probabilities, headers embedded in data rows) while returning confident, actionable insights.

Scope assumption: 'Founders' care about four things — pipeline health, sector performance, revenue/cash flow, and team efficiency. The tool suite is designed around exactly these four concerns rather than building a generic SQL-style interface.

2. Technology Stack Decisions

Layer	Choice	Rationale
LLM	Claude claude-sonnet-4-6	Native tool-calling support; strong at structured reasoning over tabular data; available v
Backend	Python (no separate server)	Streamlit hosts agent logic directly; eliminates API server overhead within 6-hour timelin
Frontend	Streamlit	Production-quality chat UI in ~100 lines; built-in session state; deploy to Render in one
Monday API	REST/GraphQL (not MCP)	MCP adds complexity with no functional advantage for this scope; REST is fully suffici
Data prep	openpyxl + csv	Zero-dependency approach; handles the Excel quirks (merged cells, row-2 headers) th
Hosting	Render (free tier)	GitHub-connected auto-deploy; supports env variables; no credit card required

3. Architecture Decisions

Tool-calling over RAG: The data is structured and relatively small (500 total records). RAG/vector search would introduce semantic noise and inconsistent aggregations. Tool-calling lets Claude decide precisely which computation to run and gives full control over aggregation logic.

7 specialized tools over 1 generic query tool: A generic 'run query' tool would require Claude to generate data manipulation code at inference time — unreliable and slow. Purpose-built tools (get_pipeline_summary, get_sector_analysis, get_revenue_insights, etc.) produce deterministic, testable results and allow explicit business logic (e.g., what constitutes an 'active' deal).

Live fetch on every query: Compliant with the brief's 'no caching' requirement. Monday.com's GraphQL API supports cursor-based pagination so large boards are fully retrieved in a single tool call chain. Latency is displayed to the user as a transparency measure.

Caveat surfacing: The normalizer returns structured caveats (e.g., '12 deals had no close date') alongside every result. These are passed through to Claude's context so it can weave them into its response naturally, and also displayed separately in the UI as a data quality panel.

4. Data Quality Handling

Issue Found	Handling Approach
Work Order headers on Excel row 2	prepare_csv.py reads row 2 as header, skips row 1; produces clean CSV
'Sector/service' and 'Deal Stage' appearing as data values	Header deduplicate filter strips rows where Deal Name matches known headers
Closure Probability as text: High/Medium/Low	PROBABILITY_MAP converts to 0.75/0.50/0.25 for numeric aggregation; original text preserved
Missing Close Date on ~40% of deals	Flagged per-deal in caveats; excluded from date-based filters with explicit notification
Sector variants (e.g. 'Renewables', 'renewable', 'SEA')	SECTOR_ALIASES dict canonicalizes all variants; unknown values mapped to 'Unknown' value
Empty Execution Status on some work orders	Defaulted to 'Unknown'; logged as caveat
Masked financial values (relative, not absolute)	Treated as ordinal for ranking; absolute sums computed but labeled as 'masked/relative' in caveats

5. Assumptions Made

- 1. "Energy sector" queries are interpreted as Renewables + Powerline combined.
- 2. Deal Value is treated as a masked relative metric; absolute revenue totals are labeled as such.
- 3. An 'active deal' is one in stages A–I (pre-won) with status 'Open'.
- 4. Work order amounts are in INR excluding GST unless the user asks for inclusive values.
- 5. The agent does not require authentication — the hosted demo is public (no sensitive data since all values are masked).
- 6. Monday.com API v2 token with read scope is sufficient; write access is not needed.

6. Trade-offs & What I'd Do With More Time

What was cut: MCP integration (would add 1–2 hrs for marginal functional gain), chart/visualization rendering in Streamlit (deprioritized for reliability), per-owner detailed drill-down beyond the summary table.

With more time: Add a Redis cache with a 5-minute TTL to reduce Monday.com API load while keeping data fresh. Implement Monday.com webhooks to invalidate cache on board updates. Add Plotly charts for pipeline funnel visualization. Expand to support write operations (create/update items) for agentic task management.

Time spent on assignment: ~5.5 hours